

Refined upper bounds for representations by five or more unit fractions

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Abstract

Let $f_k(m, n)$ count the nondecreasing k -term unit-fraction representations of m/n . Uniformly for $k \geq 5$ and $m, n \in \mathbb{N}$, we prove

$$f_k(m, n) \ll_{\varepsilon} (kn)^{\varepsilon} \left(\frac{k^{4/3} n^2}{m} \right)^{(446/289)2^{k-5}}.$$

This replaces the exponent $8/5$ in the corresponding theorem of Elsholtz and Planitzer by $446/289$. As a consequence,

$$f_k(1, 1) \leq \gamma^{(223/1156+o(1))2^k},$$

where $\gamma = 1.264084735 \dots$ is the Vardi constant in the normalization specified below. The same bound therefore holds for the number $F(k)$ of strictly increasing representations. The coefficient $223/1156$ improves the coefficient $1/5$ recorded by Bloom and Elsholtz. The main step is a piecewise application of available four-term estimates while retaining the congruence condition arising from the first denominator.

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1 Introduction

Let

$$F(k) = \# \left\{ (a_1, \dots, a_k) \in \mathbb{N}^k : a_1 < \dots < a_k, \quad \sum_{i=1}^k \frac{1}{a_i} = 1 \right\}.$$

More generally, for positive integers m, n, r , with no coprimality condition on m and n , write

$$f_r(m, n) = \# \left\{ (a_1, \dots, a_r) \in \mathbb{N}^r : a_1 \leq \dots \leq a_r, \quad \frac{m}{n} = \sum_{i=1}^r \frac{1}{a_i} \right\}. \quad (1)$$

Thus f_r permits repeated denominators, whereas $F(r)$ does not, and $F(k) \leq f_k(1, 1)$.

The order of growth of these counts is a classical problem on Egyptian fractions. The survey of Bloom and Elsholtz [1] records, as a consequence of the $c = 8/5$ estimate of Elsholtz and Planitzer [4, Theorem 2] and the prefix–tail argument of Browning and Elsholtz [2, Section 4], the bound

$$f_k(1, 1) \leq \gamma^{(1/5+o(1))2^k}.$$

For the normalization of the constant, put

$$v_1 = 1, \quad v_{j+1} = v_j(v_j + 1), \quad \gamma = \lim_{j \rightarrow \infty} v_j^{2^{-j}} = 1.264084735 \dots \quad (2)$$

The roots $v_j^{2^{-j}}$ increase to γ , because the recurrence gives $v_{j+1}^{2^{-(j+1)}} = v_j^{2^{-j}}(1 + v_j^{-1})^{2^{-(j+1)}}$. Since their successive ratios are at most $2^{2^{-(j+1)}}$, their product converges; hence the sequence is bounded and the limit in (2) exists. In particular, $v_j \leq \gamma^{2^j}$ for every j . Our principal result is a uniform estimate for rational targets.

Theorem 1.1 (Uniform bound). *For every $\varepsilon > 0$, uniformly for $k \geq 5$ and $m, n \in \mathbb{N}$,*

$$f_k(m, n) \ll_{\varepsilon} (kn)^{\varepsilon} \left(\frac{k^{4/3} n^2}{m} \right)^{(446/289)2^{k-5}}.$$

This improves the exponent $8/5$ in [4, Theorem 2] to

$$\frac{446}{289} = \frac{8}{5} - \frac{82}{1445}.$$

For representations of unity, Theorem 1.1 can be combined with the prefix–tail argument to give the following stronger statement.

Corollary 1.2. *With γ defined by (2),*

$$\limsup_{k \rightarrow \infty} 2^{-k} \log_{\gamma} f_k(1, 1) \leq \frac{223}{1156}.$$

Equivalently,

$$f_k(1, 1) \leq \gamma^{(223/1156 + o(1))2^k} \quad (k \rightarrow \infty).$$

Corollary 1.3. *We have*

$$\limsup_{k \rightarrow \infty} 2^{-k} \log_{\gamma} F(k) \leq \frac{223}{1156},$$

and hence

$$F(k) \leq \gamma^{(223/1156 + o(1))2^k}.$$

The coefficient $223/1156 = 0.192906\dots$ is smaller than $1/5$. Elsholtz and Planitzer note in [4, Section 5] that summing their $n^{3/2}/m^{3/4}$ estimate over the full range of the first denominator leads to a worse five-term bound, because the exponent of m is too small. We use this estimate only on a short initial range, retain the congruence condition on the first denominator, and then switch to other alternatives from [4, Corollary 1]. To the best of our knowledge, the resulting improvement has not previously been recorded.

2 Preliminary estimates

We shall use the following uniform consequence of Elsholtz and Planitzer [4, Corollary 1]:

$$f_4(M, N) \ll_{\eta} N^{\eta} \min\{A, B, C, D\}, \tag{3}$$

where

$$\begin{aligned} A &= N^{3/2} M^{-3/4}, & B &= N^{8/5} M^{-1}, \\ C &= N^{4/3} M^{-2/3} + N^{28/17} M^{-8/5}, & D &= (N/M)^{5/3} + N^{4/3} M^{-2/3}. \end{aligned}$$

No coprimality hypothesis is imposed in (3), so it applies directly to $(M, N) = (u, N_u)$ below, whether or not u/N_u is reduced.

We also record an elementary estimate that will preserve the spacing of one residue class.

Lemma 2.1 (Progression sums). *Let $m \geq 1$, let r be a residue class modulo m , and let $1 \leq L < U$. Then*

$$\sum_{\substack{L < u \leq U \\ u \equiv r \pmod{m}}} u^{-q} \ll_q \begin{cases} L^{-q} + m^{-1} L^{1-q}, & q > 1, \\ L^{-q} + m^{-1} U^{1-q}, & 0 < q < 1. \end{cases}$$

For $0 < q < 1$ one also has

$$\sum_{\substack{0 < u \leq U \\ u \equiv r \pmod{m}}} u^{-q} \ll_q 1 + \frac{U^{1-q}}{m}.$$

Proof. Separate the least positive term of the progression in the interval and compare the remaining terms, which are spaced by m , with the corresponding integral. $\square$

3 The five-term estimate

The saving in Theorem 1.1 enters through the next proposition.

Proposition 3.1. *Put $X = n^2/m$ and $p = 446/289$. Uniformly for $m, n \in \mathbb{N}$,*

$$f_5(m, n) \ll_{\varepsilon} n^{\varepsilon} \left(X^{3/2} + \frac{X^p}{m} \right). \quad (4)$$

In particular,

$$f_5(m, n) \ll_{\varepsilon} n^{\varepsilon} X^p. \quad (5)$$

Proof. For a representation counted by $f_5(m, n)$, define

$$u = ma_1 - n, \quad N_u = na_1 = \frac{n(n+u)}{m}.$$

Since $n/m < a_1 \leq 5n/m$, we have

$$0 < u \leq 4n, \quad u \equiv -n \pmod{m}, \quad X < N_u \leq 5X.$$

The remaining four fractions represent u/N_u . Dropping their lower bound $a_2 \geq a_1$ gives

$$f_5(m, n) \leq \sum_{\substack{0 < u \leq 4n \\ u \equiv -n \pmod{m}}} f_4(u, N_u). \quad (6)$$

Assume first that $X \geq 1$, and put

$$\alpha = \frac{50}{289}, \quad \beta = \frac{5}{17}.$$

Since $m \geq 1$ and $0 < \alpha < \beta < 1/2$,

$$X^{\alpha} \leq X^{\beta} \leq n^{2\beta} = n^{10/17} \leq n \leq 4n.$$

Thus the cutoffs below lie inside the summation range in (6). They are intrinsic: α and β coincide with the first two transition exponents in [4, Corollary 2]. Equivalently, they are respectively the crossing points of A with $N^{28/17} M^{-8/5}$, and of $N^{28/17} M^{-8/5}$ with $(N/M)^{5/3}$.

For reference, Table 1 displays the effective estimates used in the three ranges and their total contributions. The calculations follow.

Table 1: The three ranges in (6); the common factor n^ε is omitted.

Range for u	Effective estimate from (3)	Contribution after summing
$0 < u \leq X^{50/289}$	$N_u^{3/2} u^{-3/4}$	$X^{3/2} + m^{-1} X^{446/289}$
$X^{50/289} < u \leq X^{5/17}$	$C \ll N_u^{28/17} u^{-8/5}$ by (8)	$X^{396/289} + m^{-1} X^{446/289}$
$X^{5/17} < u \leq 4n$	$(N_u/u)^{5/3} + N_u^{4/3} u^{-2/3}$	$X^{20/17} + m^{-1} X^{25/17} + X^{58/51} + X^{3/2} m^{-5/6}$

We apply (3) with $\eta = \varepsilon/2$. Since $N_u^\eta \leq (5n^2)^\eta \ll_\varepsilon n^\varepsilon$, this produces a common factor n^ε , which we suppress in the next three estimates. All sums below are restricted by $u \equiv -n \pmod{m}$, which we suppress from the subscripts.

For $0 < u \leq X^\alpha$, use A in (3). The final clause of Lemma 2.1 gives

$$\sum_{u \leq X^\alpha} N_u^{3/2} u^{-3/4} \ll X^{3/2} \left(1 + \frac{X^{\alpha/4}}{m} \right) = X^{3/2} + \frac{X^p}{m}, \quad (7)$$

because $3/2 + \alpha/4 = p$.

Suppose next that $X^\alpha < u \leq X^\beta$. In this range the second summand in C dominates the first, since

$$\frac{N_u^{4/3} u^{-2/3}}{N_u^{28/17} u^{-8/5}} = N_u^{-16/51} u^{14/15} \leq X^{-16/51+14\beta/15} = X^{-2/51} \leq 1. \quad (8)$$

Lemma 2.1, now with $q = 8/5$, yields

$$\sum_{X^\alpha < u \leq X^\beta} N_u^{28/17} u^{-8/5} \ll X^{28/17} \left(X^{-8\alpha/5} + \frac{X^{-3\alpha/5}}{m} \right) = X^{396/289} + \frac{X^p}{m}. \quad (9)$$

Finally, for $X^\beta < u \leq 4n$, use both terms in D . The cases $q = 5/3$ and $q = 2/3$ of Lemma 2.1 give, respectively,

$$\sum_{X^\beta < u \leq 4n} N_u^{5/3} u^{-5/3} \ll X^{20/17} + \frac{X^{25/17}}{m}, \quad (10)$$

$$\sum_{X^\beta < u \leq 4n} N_u^{4/3} u^{-2/3} \ll X^{58/51} + \frac{X^{4/3} n^{1/3}}{m} = X^{58/51} + X^{3/2} m^{-5/6}. \quad (11)$$

Empty ranges cause no difficulty.

Every term in (7)–(11) is bounded by a constant times $X^{3/2} + m^{-1} X^p$, proving (4) for $X \geq 1$. Since $p > 3/2$ and $m \geq 1$, this also proves (5) in that range.

If $f_5(m, n) > 0$, then $m \leq 5n$. When $X < 1$, this implies $n^2 < m \leq 5n$ and hence $n < 5$; the finitely many possibilities are absorbed into the implicit constants in (4) and (5). $\square$

4 Lifting and representations of unity

We use the following lifting statement.

Lemma 4.1 (Elsholtz–Planitzer [4, Lemma B]). *Suppose that some fixed $c > 1$ satisfies, for every $\varepsilon > 0$ and uniformly for $m, n \in \mathbb{N}$,*

$$f_5(m, n) \ll_\varepsilon n^\varepsilon \left(\frac{n^2}{m} \right)^c.$$

Then, for every $\varepsilon > 0$, uniformly for $k \geq 5$ and $m, n \in \mathbb{N}$,

$$f_k(m, n) \ll_\varepsilon (kn)^\varepsilon \left(\frac{k^{4/3} n^2}{m} \right)^{c2^{k-5}}. \quad (12)$$

Proof of Theorem 1.1. Proposition 3.1, in the form (5), supplies the hypothesis of Lemma 4.1 with $c = 446/289$. Its conclusion is precisely the asserted estimate. $\square$

Remark 4.2 (Direct specialization). Taking $m = n = 1$ in Theorem 1.1 gives

$$f_k(1, 1) \ll_{\varepsilon} k^{(223/3468)2^k + \varepsilon}.$$

Indeed, $(4/3)(446/289)2^{k-5} = (223/3468)2^k$. This improves the coefficient $1/15$ obtained from [4, Corollary 3].

Proof of Corollary 1.2. Set $p = 446/289$ and fix $L \geq 5$. For $k \geq L$, let $J = k - L$ and consider a representation counted by $f_k(1, 1)$. The classical Curtiss bound [3], in the form used in [2, Section 4], states that after j terms its positive residual R_j satisfies, for $0 \leq j < k$,

$$R_j \geq \frac{1}{v_{j+1}}. \quad (13)$$

The non-strict inequality in (13) is intentional. For $1 \leq j \leq k$, we have $R_{j-1} \leq (k - j + 1)/a_j$, and hence

$$a_j \leq (k - j + 1)v_j \leq kv_j \leq k\gamma^{2^j}. \quad (14)$$

Classify the representations counted by $f_k(1, 1)$ according to their first J denominators. By (14), the number of possible prefixes and the product of the denominators in any one prefix are both at most

$$Q := k^J \gamma^{\sum_{j=1}^J 2^j} = k^J \gamma^{2^{J+1}-2}. \quad (15)$$

Write the residual after a fixed prefix in lowest terms as m/n . Its denominator divides the product of the prefix denominators, so $n \leq Q$. Moreover, (13) gives

$$\frac{n}{m} = \frac{1}{R_J} \leq v_{J+1} \leq \gamma^{2^{J+1}}.$$

Consequently

$$\frac{n^2}{m} = n \frac{n}{m} \leq k^J \gamma^{2^{J+2}-2}. \quad (16)$$

For each prefix, dropping the lower bound $a_{J+1} \geq a_J$ on the remaining denominators bounds the number of tails by $f_L(m, n)$. Apply Theorem 1.1 with $k = L$ and $\varepsilon = \delta$, use $n \leq Q$ and (16), and finally multiply by the prefix count in (15). Here and below L is fixed when k tends to infinity. Taking logarithms to base γ gives

$$\log_{\gamma} f_k(1, 1) \leq p2^{k-3} + (1 + \delta)2^{k-L+1} + O_{L,\delta}(k \log k). \quad (17)$$

For fixed L and $\delta > 0$, division by 2^k and passage to the upper limit therefore yield

$$\limsup_{k \rightarrow \infty} 2^{-k} \log_{\gamma} f_k(1, 1) \leq \frac{p}{8} + (1 + \delta)2^{1-L}.$$

Letting $L \rightarrow \infty$ proves the corollary, since

$$\frac{p}{8} = \frac{446}{8 \cdot 289} = \frac{223}{1156}.$$

$\square$

Proof of Corollary 1.3. Starting with $1 = 1/2 + 1/3 + 1/6$, split the largest denominator successively by $1/a = 1/(a+1) + 1/[a(a+1)]$. This shows that $F(k) > 0$ for every $k \geq 3$. Every strictly increasing representation counted by $F(k)$ is counted by $f_k(1, 1)$. Thus $F(k) \leq f_k(1, 1)$, and the result follows from Corollary 1.2. $\square$

Remark 4.3 (Normalization). Elsholtz and Planitzer [4, Corollary 3] use the shifted sequence beginning at index 0, whose limit is $\gamma^2 = 1.597910\dots$, whereas [1] uses the normalization (2). Thus numerical coefficients should be compared only after converting the base. Explicitly,

$$\gamma^{(223/1156+o(1))2^k} = (\gamma^2)^{(223/1156+o(1))2^{k-1}} = (\gamma^2)^{(223/2312+o(1))2^k}.$$

In the present normalization, the baseline coefficient $1/5$ follows directly by replacing p by $8/5$ in the calculation leading to (17).

5 Concluding remarks

The gain comes from two choices made before the lifting step: we retain the congruence $u \equiv -n \pmod{m}$ in (6), and we switch among the A, C, D alternatives in (3). The congruence spacing is essential for the exponent p in the third range. Without it, the second term of D would give only

$$X^{4/3}n^{1/3} = X^{3/2}m^{1/6} \ll X^{5/3},$$

where $f_5(m, n) > 0$ implies $m \leq 5n$ and hence $m \leq 25X$. This is too large because $p < 5/3$. The first two ranges already give $O(X^p)$ without progression spacing, although the stronger m -dependent terms in (4) still use it.

The critical transition is $u \asymp X^{50/289}$, where, apart from the common n^ε factor, the available pointwise estimate contributes $X^{396/289}$ for each admissible u . The discarded condition $a_2 \geq a_1$ gives no further saving there: throughout $u \leq X^\beta \leq n$, every remaining denominator b satisfies

$$b > \frac{N_u}{u} = \frac{n}{u}a_1 \geq a_1.$$

Thus any uniform power saving on the critical range must sharpen the available four-term input or exploit additional arithmetic structure; it would feed directly into a smaller exponent in Theorem 1.1 and Corollary 1.2.

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